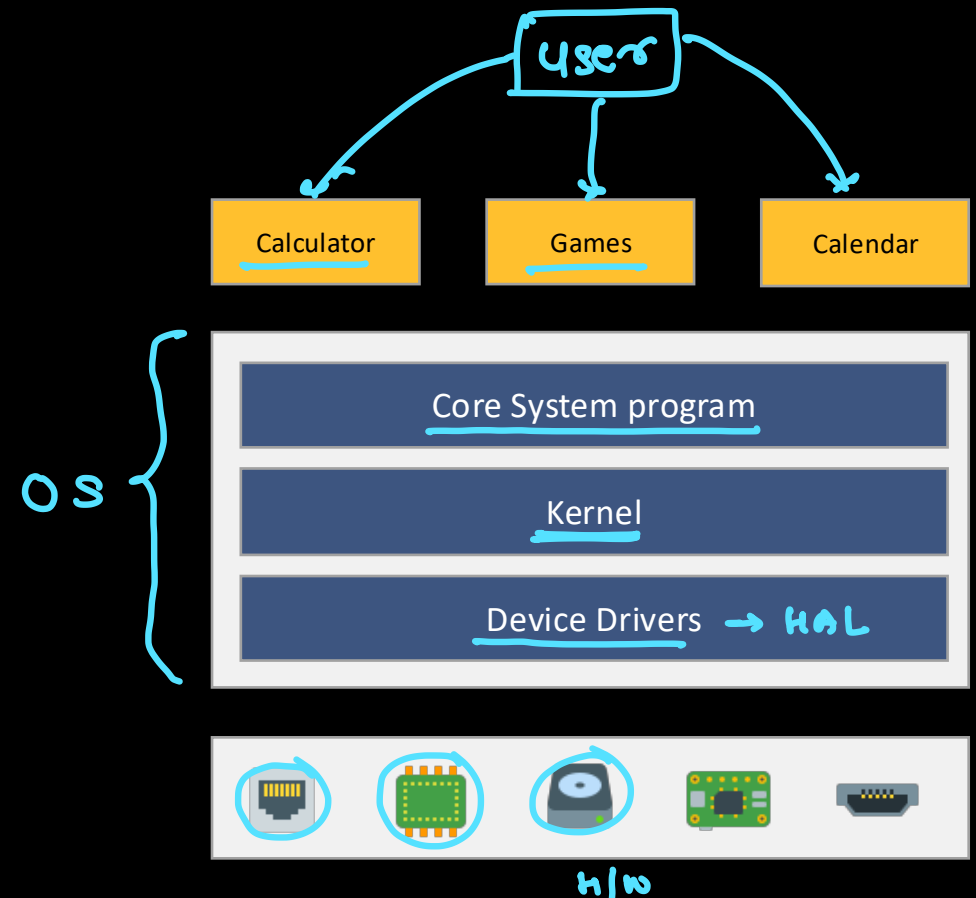




Mobile Development

What is Operating System ?

- An operating system is a system software that manages computer hardware, software resources and provides common services for computer program
- Responsibilities
 - Memory management
 - Process management
 - File System management
 - CPU Scheduling
 - Hardware abstraction



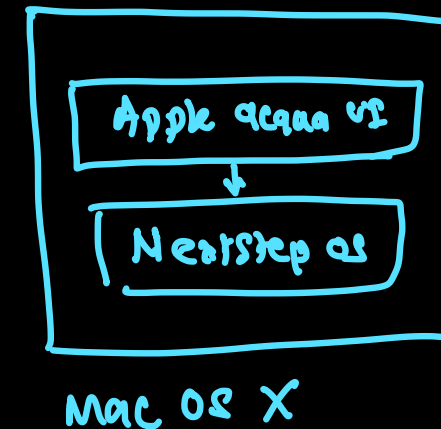
Unix

Unix has Flavors, Unix has distros



- Unix is one of the oldest operating systems still in use
- It was created in 1969 by Ken Thompson and Dennis Ritchie
- It was not released as open-source software, instead, Unix versions were associated with many different tech organizations, including IBM, Hewlett-Packard, and AT&T
- These various Unix versions are referred to as Unix "flavors" and were proprietary to each company
- E.g.
 - HP-UX → HP
 - SunOS and Solaris → SUN
 - IRIX → Silicon Graphics (openGL)
 - AIX → IBM
 - Digital Unix → Digital Software
 - BSD → Berkely Software Distribution
 - SCO Unix → SCOware
 - NeXTSTEP and OpenStep → NextStep Inc.
 - Apple macOS → **

freeBSD **
NetBSD
OpenBSD



BSD Unix



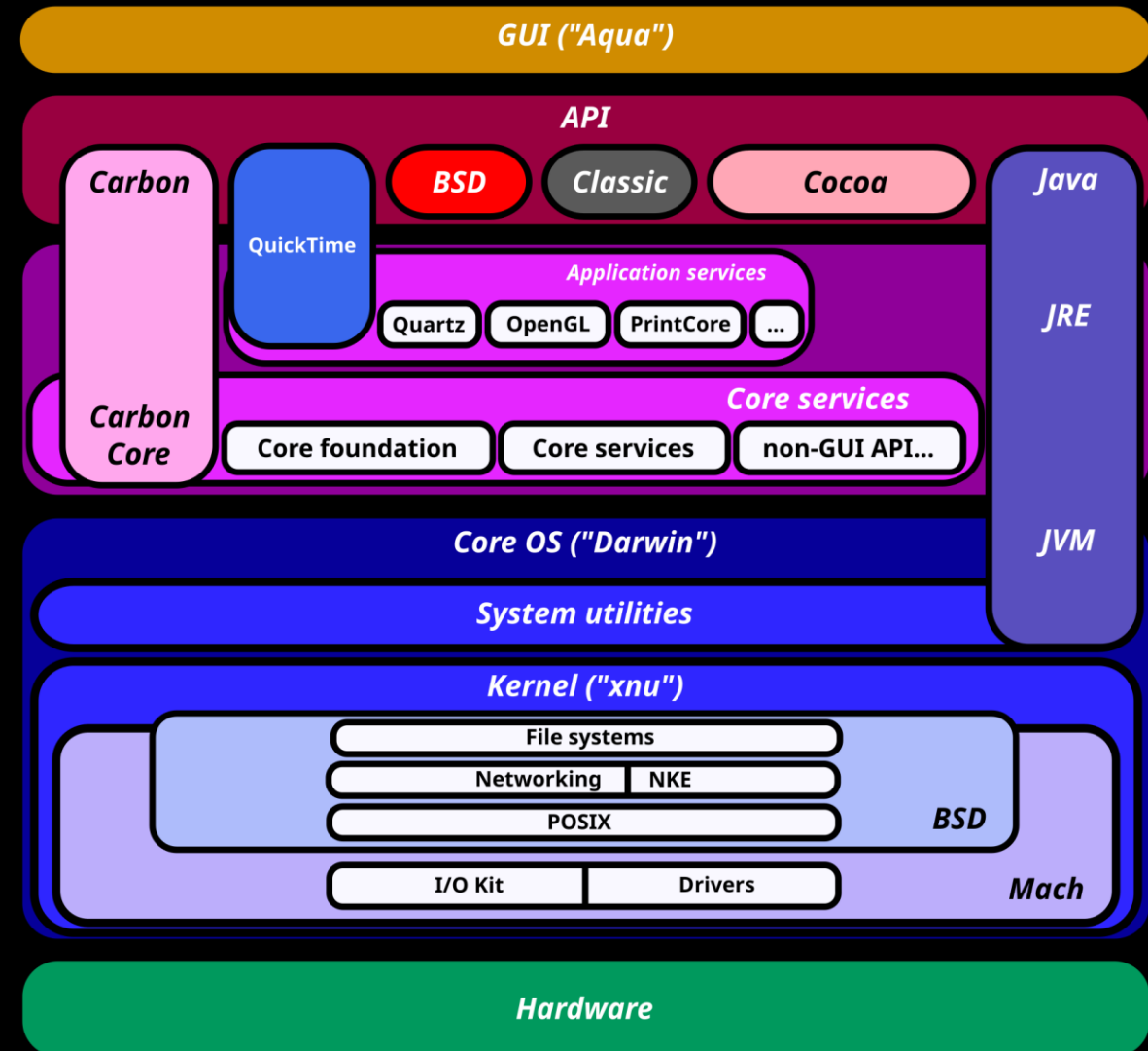
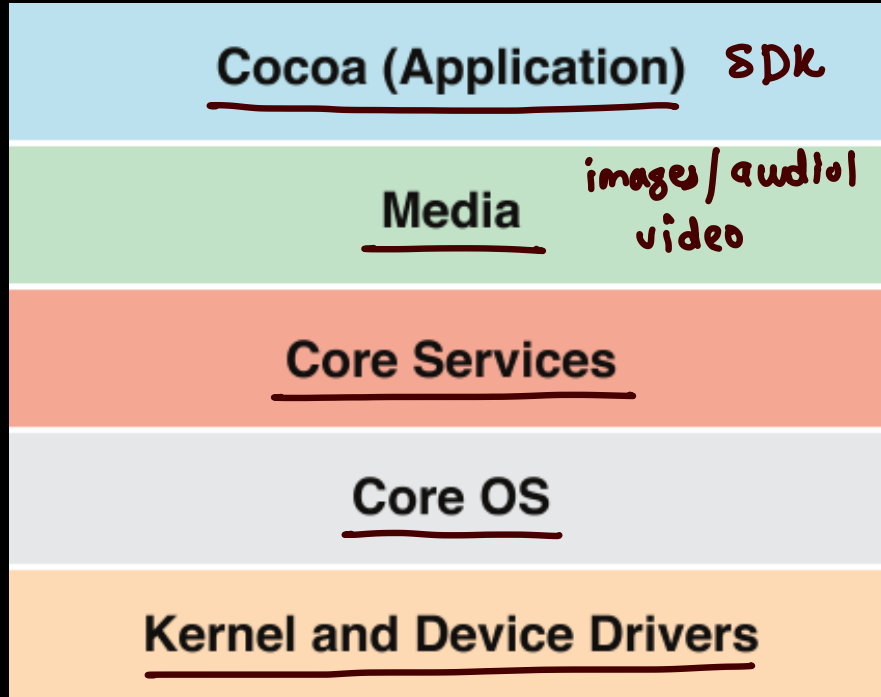
- BSD (Berkeley Software Distribution) introduced several important features and enhancements to Unix, many of which have influenced modern operating systems. Here are some key contributions:
 - **Networking:** BSD was instrumental in developing TCP/IP networking protocols, which became the standard for internet communication. → *Socket, pipe*
 - **File System Improvements:** It introduced the Fast File System (FFS), which improved performance and reliability of file storage.
 - **Virtual Memory:** BSD incorporated advanced virtual memory management techniques, enhancing memory efficiency and multitasking capabilities.
 - **User Utilities:** BSD included many user commands and utilities, such as tcsh (an enhanced shell) and tools for networking and system administration.
 - **Development Tools:** It provided a comprehensive set of programming tools, including the **make** utility for automating builds.
 - **Security Features:** Over time, BSD has included various security enhancements, such as Access Control Lists (ACLs) and improved user authentication methods.
 - **Modular Kernel:** BSD's approach to a **modular kernel** architecture allowed for easier customization and extension.
↳ *micro kernel*

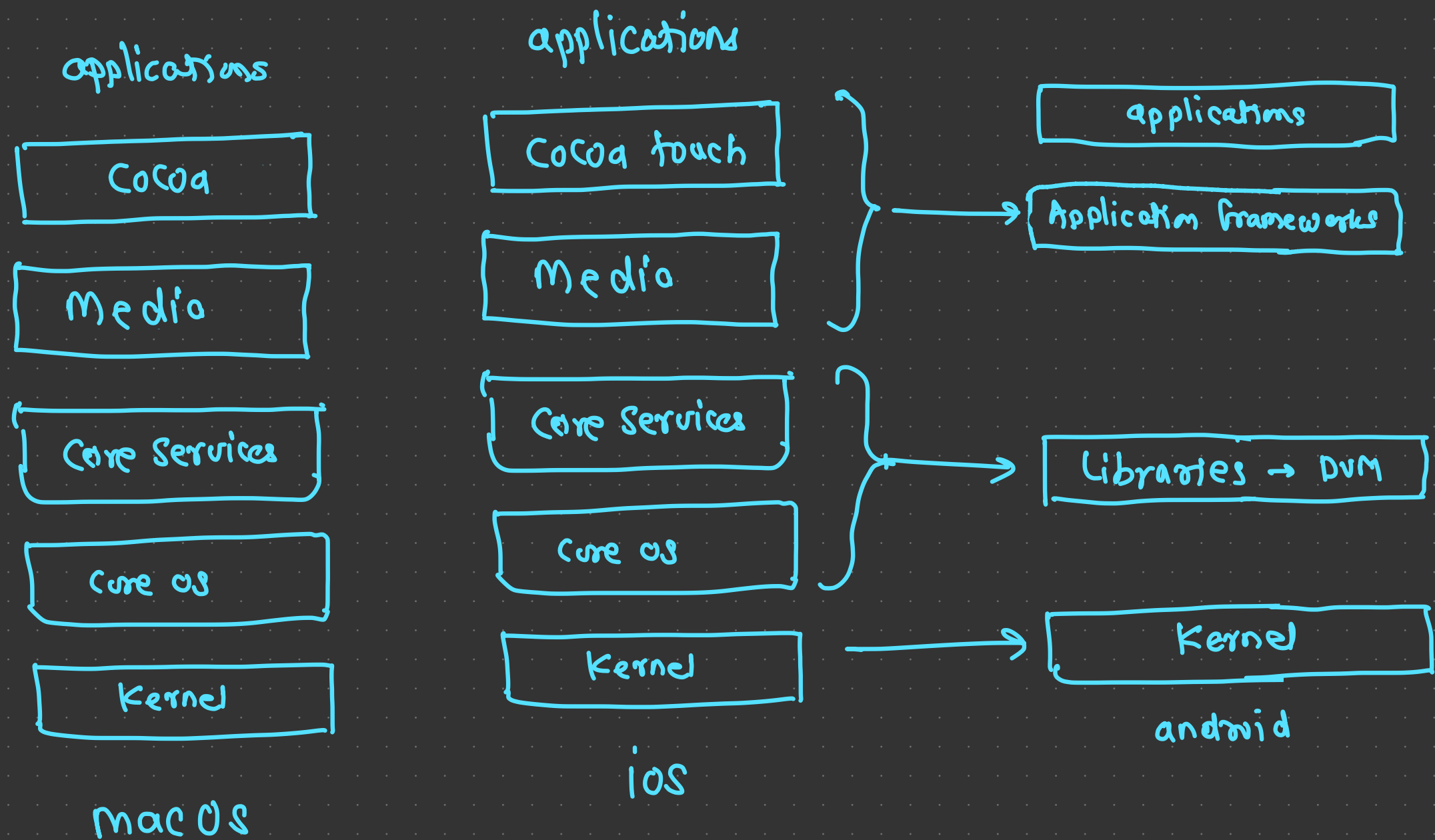
macOS

Macintosh

- macOS is Apple's operating system for Mac computers. It provides the user interface and acts as the link between a Mac's hardware and its software
- It is based on Unix technology developed at NeXT, the company Steve Jobs founded before returning to Apple
- macOS is built on a hybrid kernel called XNU, which combines components from the Mach microkernel and BSD (Berkeley Software Distribution)
- **Mach:** Provides fundamental abstractions such as tasks and threads, and handles Inter-Process Communication (IPC) through "ports"
- **BSD:** Implements the system call interface, process management, file permissions, and the file system (now APFS). It also includes key security frameworks like KAuth (Kernel Authorization) and MACF (Mandatory Access Control Framework)
- **Security and Boot Process:**
 - Apple employs a defense-in-depth approach to security
 - Starting with hardware, each stage of the boot chain validates the next
 - On Apple Silicon Macs, the boot security is managed by iBoot and the Secure Enclave
 - The "Startup Security Utility" in RecoveryOS allows administrators to configure security policies: Full Security (most restrictive), Reduced Security (allows downgrades and notarized kernel extensions), and Permissive Security (allows custom kernels, SIP must be disabled)
 - On Intel Macs, similar security levels (Full, Medium, No Security) can be set in the Startup Security Utility

macOS Architecture





Features of macOS

user interfaces → Linux → GNOME, XFCE, KDE, Cinamon
→ windows → Aero, Metro
→ macOS → Aqua

GNU



■ User Friendly Interface → Aqua

- The window and menu that makeup macOS user friendly interface which make it simple to navigate and find the information you need.

■ Built in Apps

- In macOS so many apps are pre-installed like Mail for email and Photos for picture management. similarly, safari is used for online browsing is also pre installed in it.

■ Integration with Other Apple Devices

- macOS connects easily with other Apple devices, such as the iPad, iPhone, and others. Not only you can reply to messages and take calls, but you may even begin things on one device and complete them on another.

■ Privacy and Security

- macOS is designed with security and protecting your data from malware and viruses. By allowing you to choose which apps can access your data.

■ Updates

- By apple, New features like enhancements and security patches are frequently added to macOS through regular updates. These updates keep your Mac operating properly and are generally simple to apply.

■ Accessibility

- macOS includes features like VoiceOver for visually disabled users and Dictation for individuals who have trouble in typing.

Advantages of macOS



- **Security**
 - macOS is less vulnerable to malware and security problems than other operating systems.
- **Integration with Apple Devices**
 - macOS has good interoperability with other Apple products, such as iPads and iPhones, enabling smooth tasks and data transfer.
- **Multitasking**
 - This enables users to quickly and effortlessly switch between applications and complete tasks.
- **Free Productivity Applications** → XCode
 - macOS is pre installed with a number of useful free productivity apps.
- **Long Life**
 - macOS has been highly durable, with certain versions still receiving support years after they were first released.
- **Good Performance**
 - Fast boot speeds and seamless operation, even on outdated hardware, characterize Mac OS performance.

Disadvantages of macOS



- **High Cost**

- Compared to comparable Windows-based solutions, Mac goods are typically more expensive, which can be a deterrent for many customers.

- **Limited Software Availability**

- Some users may find it aggravating since Mac OS supports fewer games and software programs than Windows.

- **Lack of Hardware Customization**

- Those who want a more customized setup may find that macOS offers less choices for customizing hardware.

- **Incompatible File System**

- Moving files between the two operating systems is challenging because the macOS file system is incompatible with Windows.

- **Lack of Flexibility for Hardware Upgrades**

- It may be challenging to upgrade or fix Mac based systems due to the restricted alternatives for hardware upgrades.

macOS versions



- Mac OS X 10.0 (Cheetah) - March 24, 2001
- Mac OS X 10.1 (Puma) - September 25, 2001
- Mac OS X 10.2 (Jaguar) - August 23, 2002
- Mac OS X 10.3 (Panther) - October 24, 2003
- Mac OS X 10.4 (Tiger) - April 29, 2005
- Mac OS X 10.5 (Leopard) - October 26, 2007
- Mac OS X 10.6 (Snow Leopard) - August 28, 2009
- Mac OS X 10.7 (Lion) - July 20, 2011
- OS X 10.8 (Mountain Lion) - July 25, 2012
- OS X 10.9 (Mavericks) - October 22, 2013
- OS X 10.10 (Yosemite) - October 16, 2014
- OS X 10.11 (El Capitan) - September 30, 2015

macOS Versions



- macOS 10.12 (Sierra) - September 20, 2016
- macOS 10.13 (High Sierra) - September 25, 2017
- macOS 10.14 (Mojave) - September 24, 2018
- macOS 10.15 (Catalina) - October 7, 2019
- macOS 11 (Big Sur) - November 12, 2020
- macOS 12 (Monterey) - October 25, 2021
- macOS 13 (Ventura) - October 24, 2022
- macOS 14 (Sonoma) - September 26, 2023
- macOS 15 (Sequoia) - September 16, 2024 → base
- macOS 16 (Tahoe) - June 9, 2025 (announced)

Apple Hardware Lineup



■ Mac Lineup

■ Laptops

- MacBook Air (M2 / M3) – 13-inch and 15-inch
- MacBook Pro (M3 Pro / M3 Max) – 14-inch and 16-inch

■ Desktops

- Mac mini (M2 / M2 Pro)
- Mac Studio (M2 Max / M2 Ultra)
- Mac Pro (Apple Silicon, M2 Ultra) – Tower and Rack versions
- iMac (M3) – 24-inch all-in-one

■ iPhone Lineup

■ Main Series

- iPhone 15 / 15 Plus
- iPhone 15 Pro / 15 Pro Max

■ Budget Model

- iPhone SE (3rd generation)

Apple Hardware Lineup



■ iPad Lineup

- iPad (10th generation) – Base model
- iPad mini (6th generation)
- iPad Air (M2)
- iPad Pro (M4) – 11-inch and 13-inch, OLED display

■ Apple Watch Lineup

- Apple Watch Series 9
- Apple Watch SE (2nd generation)
- Apple Watch Ultra 2

■ Audio Products

- AirPods (2nd and 3rd generation) →
- AirPods Pro (2nd generation with USB-C)
- AirPods Max

■ TV and Home

- Apple TV 4K (3rd generation)
- HomePod (2nd generation)
- HomePod mini

■ Spatial Computing

- Apple Vision Pro – Mixed reality headset

Desktop → iMac, Mac mini → macOS

mobile → iPhone → iOS

Smart watch → Apple Watch → watchOS

vision → Vision VR → visionOS

TU → Apple TV → tvOS

tablet → iPad, iPad Pro → iPadOS

IDE ⇒ XCode

language → Swift



ios

iOS → stripped down MacOS



- iOS represents Apple's mobile operating system, first introduced alongside the original iPhone in 2007
- The platform revolutionized mobile computing by introducing intuitive touch interactions and a simplified user interface that eliminated the need for styluses and physical keyboards
- Unlike Android, iOS maintains a closed ecosystem where Apple controls both the hardware and software, enabling consistent performance and security across supported devices
- The operating system has evolved from a basic phone interface to a sophisticated platform that powers millions of apps across various categories, from social media to professional tools
- iOS established many conventions we now take for granted in mobile computing, such as the app store model for software distribution and multitouch gestures for navigation
- The platform's focus on privacy and security has become increasingly central to its identity, with features like App Tracking Transparency fundamentally changing how mobile apps handle user data
- iOS shares its core foundation with iPadOS, though the two systems have diverged to better serve their respective device categories
- The operating system's integration with other Apple services and devices creates a seamless experience across the company's entire product lineup
- Each annual release of iOS brings new capabilities while maintaining the platform's reputation for stability and ease of use.
- The system continues to shape the mobile industry through innovations in areas like augmented reality, on-device machine learning, and user privacy protection.

iOS Architecture

Core OS Layer (Lowest Level)

- Contains the kernel (XNU kernel based on Mach)
- Manages low-level features like memory management, file system, security
- Handles networking protocols, system drivers
- Includes frameworks like Core Bluetooth and Security Services
- Manages power and battery management

Core Services Layer

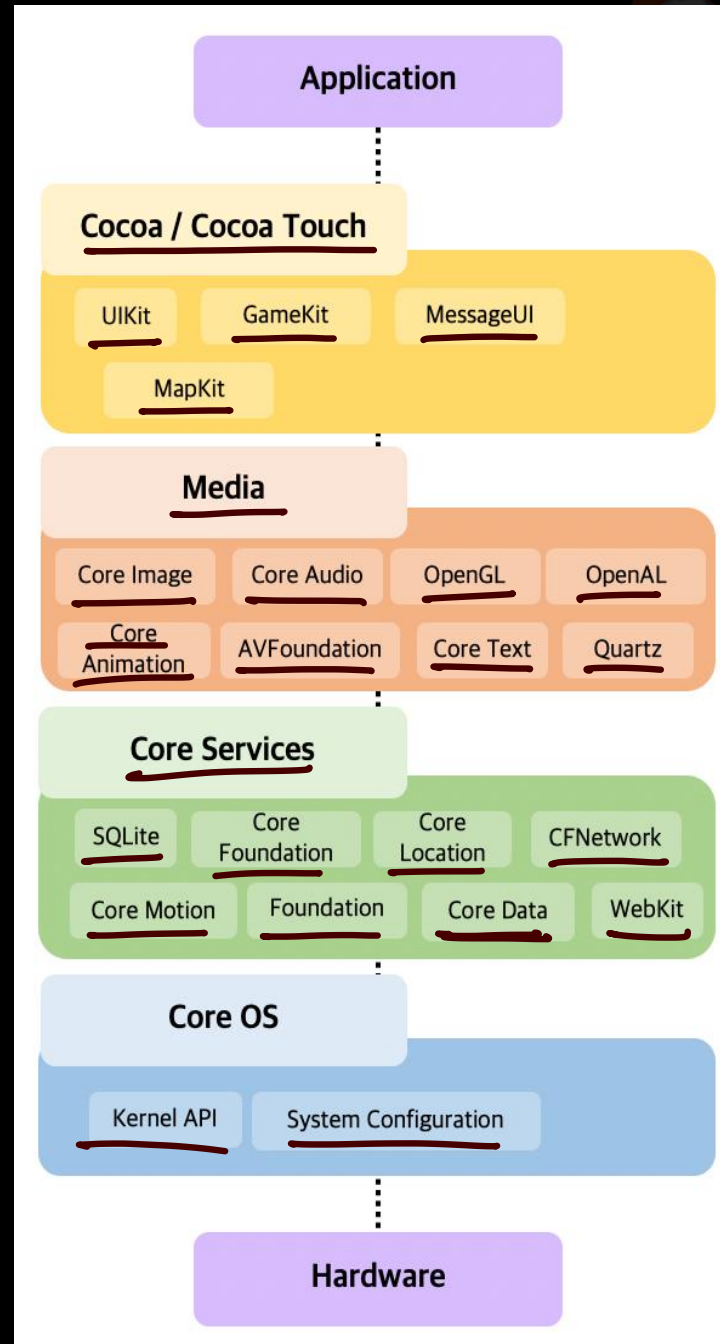
- Handles fundamental system services and technologies
- Contains Core Foundation and Foundation frameworks
- Manages data storage and iCloud
- Handles location services, networking features
- SQLite database support → Core Data
- XML parsing and threading support

Media Layer

- Graphics and audio processing
- Core Audio, OpenGL ES, Metal for graphics
- Video playback and recording
- Core Animation for fluid UI animations
- Image file handling and processing
- PDF rendering

Cocoa Touch Layer (Highest Level)

- Direct user interaction layer
- UIKit framework for UI elements
- View hierarchy and event handling
- Touch and gesture recognition
- Multitasking support
- Push notifications → APNS = FCM
- High-level networking

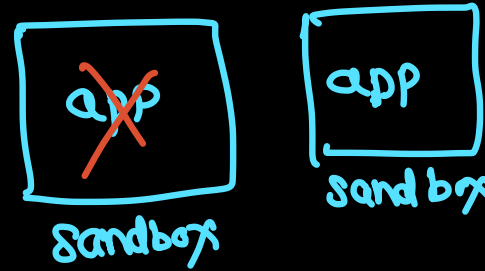


Features of iOS



▪ Sandboxing

- Each app runs in its own protected space
- Limited access to system resources
- Enhanced security and stability



▪ Memory Management

- Automatic Reference Counting (ARC) = Garbage collection
- Virtual memory system
- App state management

▪ Security Model

- Code signing requirements = digital certification
- App Store review process
- Permission-based access to device features
- Data encryption

▪ Communication Patterns

- Inter-process communication through XPC
- Delegate pattern for object communication
- Notification system for broadcast events
- Key-Value Observing for property changes

iOS Versions



▪ Early iPhone OS Versions

- iPhone OS 1.0 - June 29, 2007
- iPhone OS 2.0 - July 11, 2008
- iPhone OS 3.0 - June 17, 2009

} No multi processing

▪ iOS Versions

- iOS 4 - June 21, 2010
- iOS 5 - October 12, 2011
- iOS 6 - September 19, 2012
- iOS 7 - September 18, 2013
- iOS 8 - September 17, 2014
- iOS 9 - September 16, 2015
- iOS 10 - September 13, 2016
- iOS 11 - September 19, 2017
- iOS 12 - September 17, 2018
- iOS 13 - September 19, 2019
- iOS 14 - September 16, 2020
- iOS 15 - September 20, 2021
- iOS 16 - September 12, 2022
- iOS 17 - September 18, 2023
- iOS 18 - September 16, 2024
- iOS 26 - September 15, 2025



iPhone





First Generation

- iPhone (2G) - June 29, 2007
 - Initial OS: iPhone OS 1.0
 - Screen Size: 3.5 inches
 - Resolution: 320 x 480 pixels (165 ppi)



3G Series



- **iPhone 3G - July 11, 2008**
 - Initial OS: iPhone OS 2.0
 - Screen Size: 3.5 inches
 - Resolution: 320 x 480 pixels (165 ppi)
- **iPhone 3GS - June 19, 2009**
 - Initial OS: iPhone OS 3.0
 - Screen Size: 3.5 inches
 - Resolution: 320 x 480 pixels (165 ppi)



4 Series

- iPhone 4 - June 24, 2010
 - Initial OS: iOS 4.0
 - Screen Size: 3.5 inches
 - Resolution: 640 x 960 pixels (326 ppi) - Retina Display
- iPhone 4S - October 14, 2011
 - Initial OS: iOS 5.0
 - Screen Size: 3.5 inches
 - Resolution: 640 x 960 pixels (326 ppi) - Retina Display



5 Series

- iPhone 5 - September 21, 2012
 - Initial OS: iOS 6.0
 - Screen Size: 4.0 inches
 - Resolution: 640 x 1136 pixels (326 ppi)
- iPhone 5S - September 20, 2013
 - Initial OS: iOS 7.0
 - Screen Size: 4.0 inches
 - Resolution: 640 x 1136 pixels (326 ppi)
- iPhone 5C - September 20, 2013
 - Initial OS: iOS 7.0
 - Screen Size: 4.0 inches
 - Resolution: 640 x 1136 pixels (326 ppi)



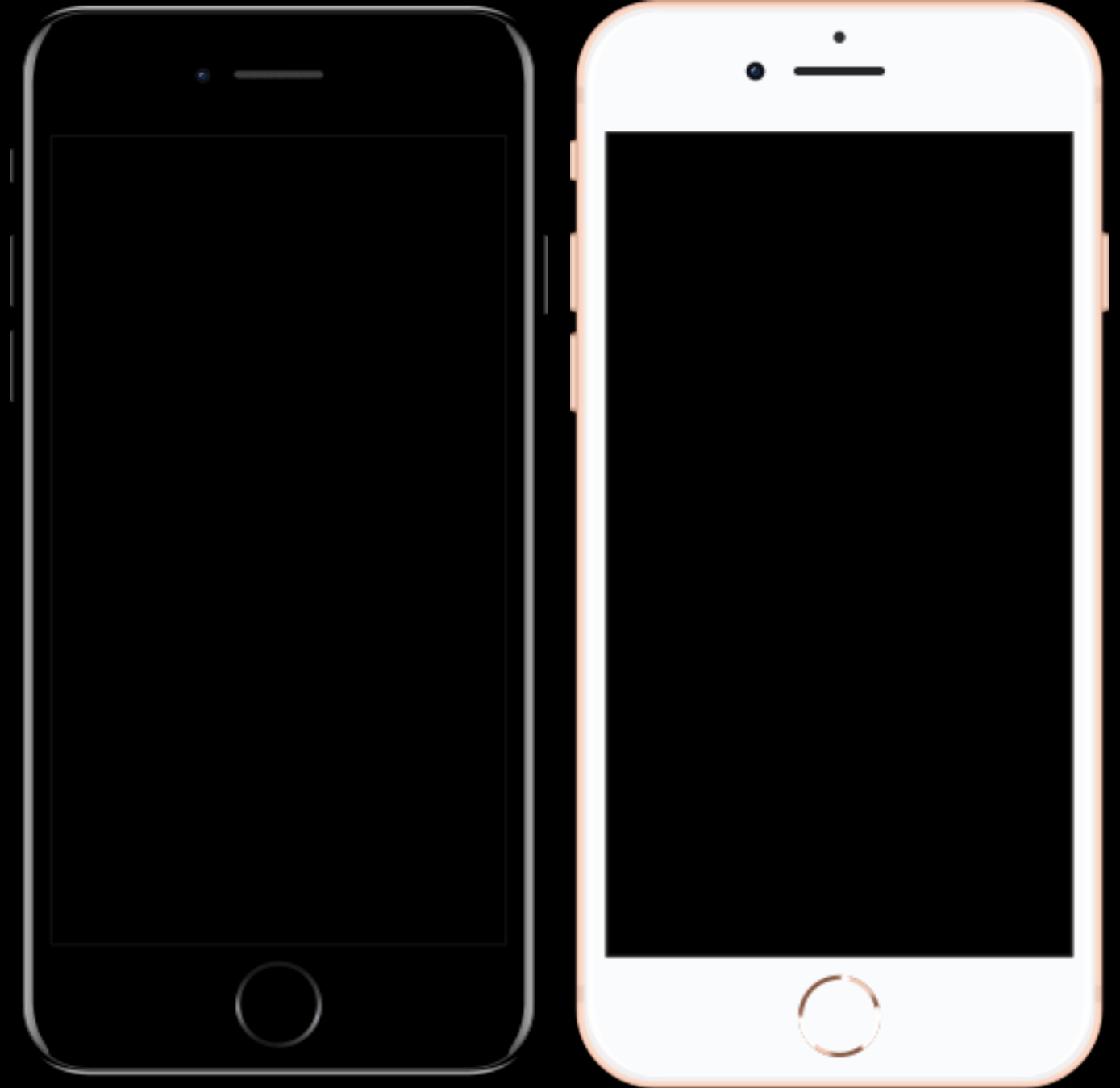
6 Series

- iPhone 6 - September 19, 2014
 - Initial OS: iOS 8.0
 - Screen Size: 4.7 inches
 - Resolution: 750 x 1334 pixels (326 ppi)
- iPhone 6 Plus - September 19, 2014
 - Initial OS: iOS 8.0
 - Screen Size: 5.5 inches
 - Resolution: 1080 x 1920 pixels (401 ppi)
- iPhone 6S - September 25, 2015
 - Initial OS: iOS 9.0
 - Screen Size: 4.7 inches
 - Resolution: 750 x 1334 pixels (326 ppi)
- iPhone 6S Plus - September 25, 2015
 - Initial OS: iOS 9.0
 - Screen Size: 5.5 inches
 - Resolution: 1080 x 1920 pixels (401 ppi)



SE and 7 Series

- **iPhone SE (1st generation) - March 31, 2016**
 - Initial OS: iOS 9.3
 - Screen Size: 4.0 inches
 - Resolution: 640 x 1136 pixels (326 ppi)
- **iPhone 7 - September 16, 2016**
 - Initial OS: iOS 10.0
 - Screen Size: 4.7 inches
 - Resolution: 750 x 1334 pixels (326 ppi)
- **iPhone 7 Plus - September 16, 2016**
 - Initial OS: iOS 10.0
 - Screen Size: 5.5 inches
 - Resolution: 1080 x 1920 pixels (401 ppi)



8 and X Series

- iPhone 8 - September 22, 2017

- Initial OS: iOS 11.0
- Screen Size: 4.7 inches
- Resolution: 750 x 1334 pixels (326 ppi)

- iPhone 8 Plus - September 22, 2017

- Initial OS: iOS 11.0
- Screen Size: 5.5 inches
- Resolution: 1080 x 1920 pixels (401 ppi)

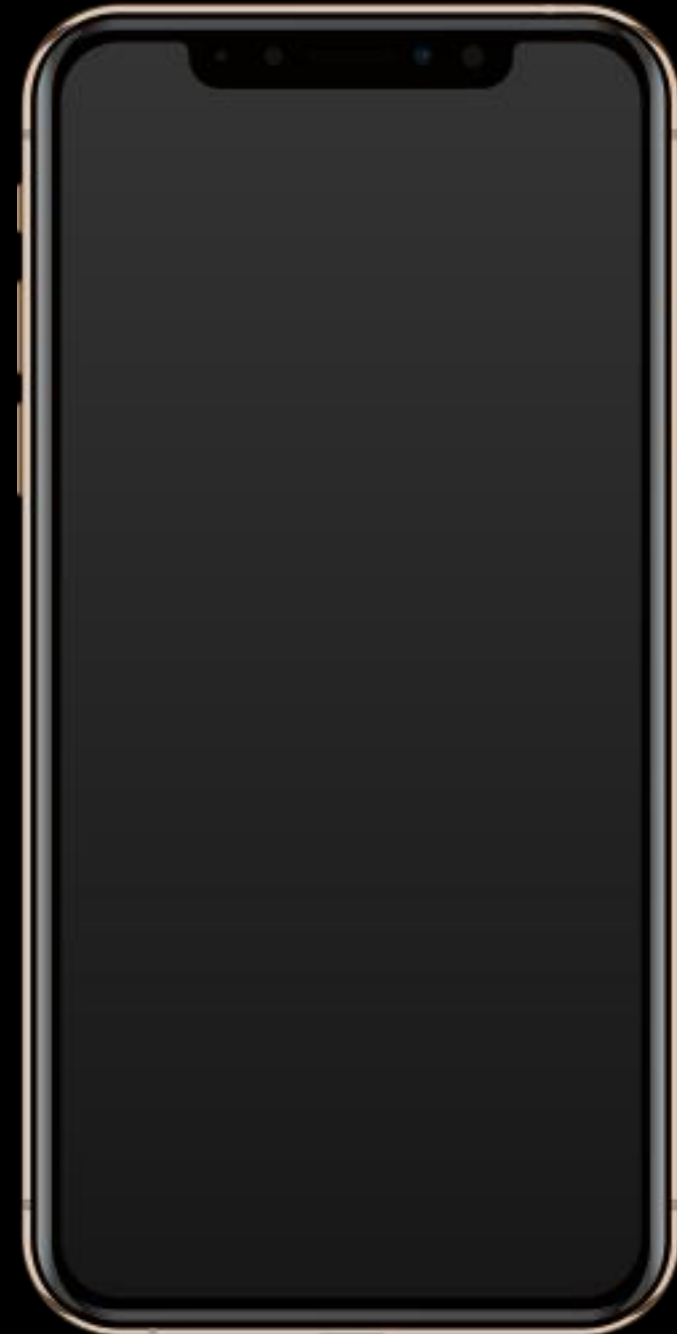
- iPhone X - November 3, 2017

- Initial OS: iOS 11.1
- Screen Size: 5.8 inches
- Resolution: 1125 x 2436 pixels (458 ppi) - Super Retina HD



XS and XR Series

- iPhone XS - September 21, 2018
 - Initial OS: iOS 12.0
 - Screen Size: 5.8 inches
 - Resolution: 1125 x 2436 pixels (458 ppi) - Super Retina HD
- iPhone XS Max - September 21, 2018
 - Initial OS: iOS 12.0
 - Screen Size: 6.5 inches
 - Resolution: 1242 x 2688 pixels (458 ppi) - Super Retina HD
- iPhone XR - October 26, 2018
 - Initial OS: iOS 12.0
 - Screen Size: 6.1 inches
 - Resolution: 828 x 1792 pixels (326 ppi) - Liquid Retina HD



11 Series

- iPhone 11 - September 20, 2019

- Initial OS: iOS 13.0
- Screen Size: 6.1 inches
- Resolution: 828 x 1792 pixels (326 ppi) - Liquid Retina HD

- iPhone 11 Pro - September 20, 2019

- Initial OS: iOS 13.0
- Screen Size: 5.8 inches
- Resolution: 1125 x 2436 pixels (458 ppi) - Super Retina XDR

- iPhone 11 Pro Max - September 20, 2019

- Initial OS: iOS 13.0
- Screen Size: 6.5 inches
- Resolution: 1242 x 2688 pixels (458 ppi) - Super Retina XDR



SE 2 and 12 Series



- iPhone SE (2nd generation) - April 24, 2020

- Initial OS: iOS 13.4
- Screen Size: 4.7 inches
- Resolution: 750 x 1334 pixels (326 ppi)

- iPhone 12 Mini - November 13, 2020

- Initial OS: iOS 14.1
- Screen Size: 6.1 inches
- Resolution: 1170 x 2532 pixels (460 ppi) - Super Retina XDR

- iPhone 12 Pro - October 23, 2020

- Initial OS: iOS 14.1
- Screen Size: 6.1 inches
- Resolution: 1170 x 2532 pixels (460 ppi) - Super Retina XDR

- iPhone 12 Pro Max - November 13, 2020

- Initial OS: iOS 14.2
- Screen Size: 6.7 inches
- Resolution: 1284 x 2778 pixels (458 ppi) - Super Retina XDR



13 Series

■ iPhone 13 Mini - September 24, 2021

- Initial OS: iOS 15.0
- Screen Size: 5.4 inches
- Resolution: 1080 x 2340 pixels (476 ppi) - Super Retina XDR

■ iPhone 13 - September 24, 2021

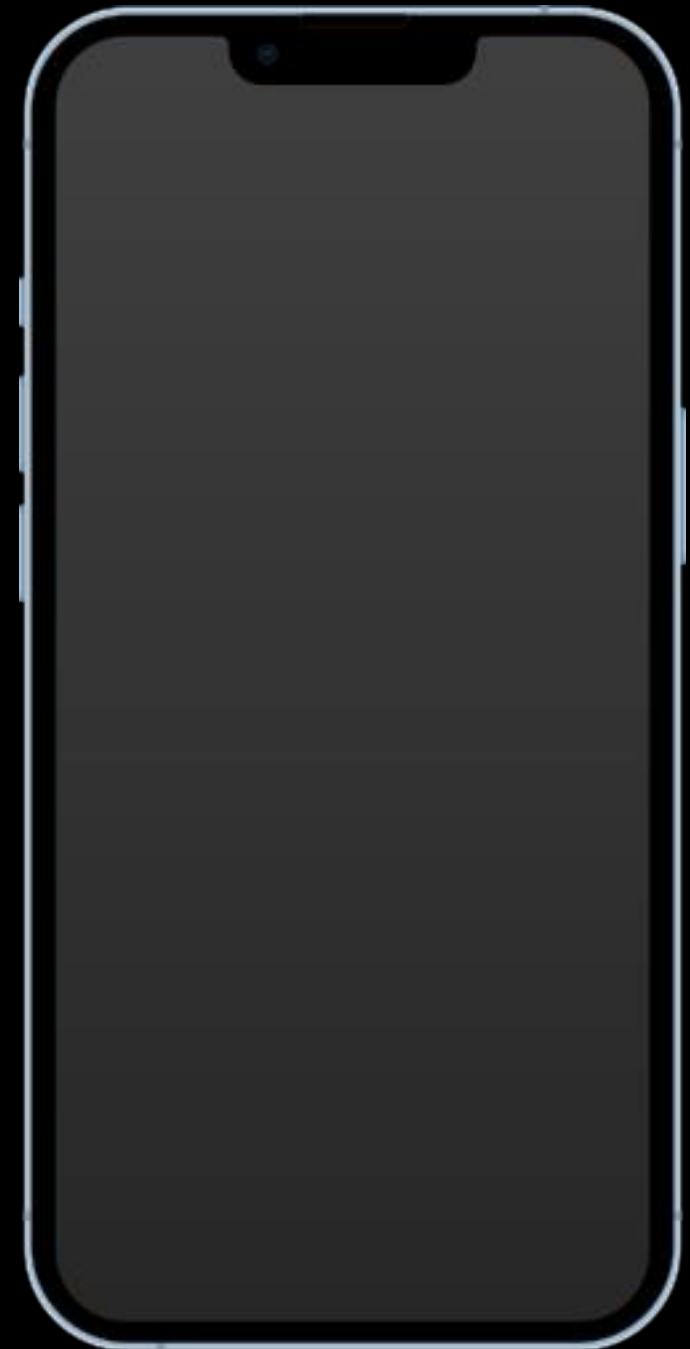
- Initial OS: iOS 15.0
- Screen Size: 6.1 inches
- Resolution: 1170 x 2532 pixels (460 ppi) - Super Retina XDR

■ iPhone 13 Pro - September 24, 2021

- Initial OS: iOS 15.0
- Screen Size: 6.1 inches
- Resolution: 1170 x 2532 pixels (460 ppi) - Super Retina XDR

■ iPhone 13 Pro Max - September 24, 2021

- Initial OS: iOS 15.0
- Screen Size: 6.7 inches
- Resolution: 1284 x 2778 pixels (458 ppi) - Super Retina XDR



SE 3 and 14 Series

- **iPhone SE (3rd generation) - March 18, 2022**

- Initial OS: iOS 15.4
- Screen Size: 4.7 inches
- Resolution: 750 x 1334 pixels (326 ppi)

- **iPhone 14 - September 16, 2022**

- Initial OS: iOS 16.0
- Screen Size: 6.1 inches
- Resolution: 1170 x 2532 pixels (460 ppi) - Super Retina XDR

- **iPhone 14 Plus - October 7, 2022**

- Initial OS: iOS 16.0
- Screen Size: 6.7 inches
- Resolution: 1284 x 2778 pixels (458 ppi) - Super Retina XDR

- **iPhone 14 Pro - September 16, 2022**

- Initial OS: iOS 16.0
- Screen Size: 6.1 inches
- Resolution: 1179 x 2556 pixels (460 ppi) - Super Retina XDR

- **iPhone 14 Pro Max - September 16, 2022**

- Initial OS: iOS 16.0
- Screen Size: 6.7 inches
- Resolution: 1290 x 2796 pixels (460 ppi) - Super Retina XDR



15 Series

- **iPhone 15 - September 22, 2023**
 - Initial OS: iOS 17.0
 - Screen Size: 6.1 inches
 - Resolution: 1179 x 2556 pixels (460 ppi) - Super Retina XDR
- **iPhone 15 Plus - September 22, 2023**
 - Initial OS: iOS 17.0
 - Screen Size: 6.7 inches
 - Resolution: 1290 x 2796 pixels (460 ppi) - Super Retina XDR
- **iPhone 15 Pro - September 22, 2023**
 - Initial OS: iOS 17.0
 - Screen Size: 6.1 inches
 - Resolution: 1179 x 2556 pixels (460 ppi) - Super Retina XDR
- **iPhone 15 Pro Max - September 22, 2023**
 - Initial OS: iOS 17.0
 - Screen Size: 6.7 inches
 - Resolution: 1290 x 2796 pixels (460 ppi) - Super Retina XDR



16 Series

■ iPhone 16 - September 20, 2024

- Initial OS: iOS 18.0
- Screen Size: 6.1 inches
- Resolution: 1179 x 2556 pixels (460 ppi) - Super Retina XDR

■ iPhone 16 Plus - September 20, 2024

- Initial OS: iOS 18.0
- Screen Size: 6.7 inches
- Resolution: 1290 x 2796 pixels (460 ppi) - Super Retina XDR

■ iPhone 16 Pro - September 20, 2024

- Initial OS: iOS 18.0
- Screen Size: 6.1 inches
- Resolution: 1179 x 2556 pixels (460 ppi) - Super Retina XDR

■ iPhone 16 Pro Max - September 20, 2024

- Initial OS: iOS 18.0
- Screen Size: 6.7 inches
- Resolution: 1290 x 2796 pixels (460 ppi) - Super Retina XDR

